

# Seunghwan Um

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## RESEARCH STATEMENT

I'm a PhD student at [Robotics Innovatory](#), Sungkyunkwan University, South Korea, under the supervision of [Prof. Hyouk Ryeol Choi](#).

My research focuses on *robotic manipulation* for logistics applications, specifically picking and placing tasks. This work spans the entire spectrum, from initial gripper design to the current development of an imitation learning pipeline. Ultimately, to achieve human-level manipulation, I am deeply interested in integrating fingertip sensors and developing learning policies that can effectively leverage them.

## EDUCATION

- |                                                                                                     |                                                          |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Sungkyunkwan University (SKKU)</b><br><i>Ph.D in Mechanical Engineering</i>                      | <i>Mar. 2022 - Present</i><br>Suwon, Republic of Korea   |
| <b>Hanyang University (ERICA)</b><br><i>B.S in Mechanical Engineering, Degree Honors: Cum Laude</i> | <i>Mar. 2018 - Feb. 2022</i><br>Ansan, Republic of Korea |

## PUBLICATIONS

J=JOURNAL, C=CONFERENCE, S=IN SUBMISSION, P=PATENT, T=THESIS

- [J.6] **PALM-Gripper: Integrated Gripper with Parallel Adaptable Mechanism for Shelf Picking in Logistics**  
Seunghwan Um, Yeong Gwang Son, Juyong Hong, Chun Soo Kim, et al. and Hyouk Ryeol Choi\*.  
*IEEE/ASME Transactions on Mechatronics (TMECH)*, 2026.
- [J.5] **Toward Reliable Bin-Picking: Collision-Aware Robotic Design and Control Strategy for Heavily Cluttered Environment**  
Seunghwan Um, Yeong Gwang Son, Jaeyoon Shim, Hyouk Ryeol Choi\*  
*IEEE Robotics and Automation Practice (RA-P)*, 2026, The paper was invited for publication.
- [J.4] **Plug-and-Play Shape Matching Module for Zero-Shot Mesh-Free Grasp Refinement on Unknown Objects**  
Juyong Hong, Yeong Gwang Son, Seunghwan Um, Hyouk Ryeol Choi\*.  
*IEEE Robotics and Automation Letters (RA-L)*, 2025.
- [J.3] **Corner-Grasp: Multi-Action Grasp Detection and Active Gripper Adaptation for Grasping in Cluttered Environments**  
Yeong Gwang Son, Seunghwan Um, Juyong Hong, Tat Hieu Bui, Hyouk Ryeol Choi\*.  
*ArXiv*, 2025. *Under Revision at Autonomous Robots*.
- [J.2] **Development of Adaptive Gripper Enhancing Power Grasp Range and Linearity**  
Issac Rhee, Chun Soo Kim, Heeyeon Jeong, Seunghwan Um, and Hyouk Ryeol Choi\* et al.  
*IEEE Access*, 2024.
- [J.1] **ReC-Gripper: A Reconfigurable Combined Suction and Fingered Gripper for Various Logistics Picking and Stowing Tasks**  
Seunghwan Um, Heeyeon Jeong, Chun Soo Kim, Issac Rhee, and Hyouk Ryeol Choi\*  
*IEEE Robotics and Automation Letters (RA-L)*, Presented in ICRA 2024.
- [C.1] **Overcoming Heavy Clutter: Utilizing the Hybrid Grasping Network and Gripper**  
Seunghwan Um, Yeong Gwang Son, Tat Hieu Bui, Ho Sang Jung, and Hyouk Ryeol Choi\*  
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024  
*Workshop: Benchmarking via Competitions in Robotic Grasping and Manipulation 🏆 [Best Extended Abstract]*
- [S.3] **Mesh-Free Dexterous Robot Demonstrations from Human RGB-D Video and Teleoperation**  
Juyong Hong, Seunghwan Um, Hyouk Ryeol Choi\*.  
*Submitted to IEEE Access, Under Review.*
- [S.2] **Pointwise Uncertainty-Aware Fusion of LiDAR and Radar for Dynamic Occupancy Grid Mapping and Navigation**  
Hyunchang Kang, Minh Thang Bui, Sung-Jin Han, Youngbin Song, Jaeyun Sim, Seunghwan Um, Hyouk Ryeol Choi\*.  
*Submitted to IEEE Robotics and Automation Letters (RA-L), Under Review.*
- [S.1] **Development of an Ultra-High Holding Force Density Halbach-Electro-Permanent Magnet Gripper for Robotic Applications**  
Heeyeon Jeong, Seunghwan Um, Hyunchang Kang, Jaeyun Sim, Hyouk Ryeol Choi\*.  
*Submitted to IEEE Robotics and Automation Letters (RA-L), Under Review.*

## PATENTS

- [P.6] **Horizontal Adaptive Hybrid Gripper and Robot Including the Same.**  
Seunghwan Um, Chun Soo Kim, Issac Rhee, Heeyeon Jeong, and Yoon Haeng Lee.  
Korean Intellectual Property Office, Application No. 10-2024-0131112. Application Date: 2024.09.26.
- [P.5] **Gripper Including Multi-Stage Translational Mechanism.**  
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.  
Korean Intellectual Property Office, Application No. 10-2023-0125840. Application Date: 2023.09.20.
- [P.4] **Gripper Including Telescopic Mechanism.**  
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.  
Korean Intellectual Property Office, Application No. 10-2023-0124012. Application Date: 2023.09.18.
- [P.3] **Suction Gripper Capable of Translational and Rotational Movements.**  
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.  
Korean Intellectual Property Office, Patent No. 10-2023-0093340. Registration Date: 2023.10.18.
- [P.2] **Adaptive Gripper Capable of Parallel Motion.**  
Issac Rhee, Chun Soo Kim, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.  
Korean Intellectual Property Office, Patent No. 10-2023-0077512. Registration Date: 2023.12.06.
- [P.1] **Hybrid Gripper Capable of Bin Picking and Shelf Picking.**  
Seunghwan Um, Heeyeon Jeong, Chun Soo Kim, Issac Rhee, and Yoon Haeng Lee.  
Korean Intellectual Property Office, Patent No. 10-2023-0076248. Publication Date: 2024.12.23.

## HONORS AND AWARDS

- 🏆 **Robot Engineering and Design Show (RED Show)** Feb. 2026  
Korea Robotics Society Annual Conference (KRoC) [KRoC RED Show 🌐]
- 🏆 **Samsung Humantech Paper Award** Feb. 2025  
Samsung Electronics Co., Ltd. [Samsung Humantech 🌐]
- 🏆 **9<sup>th</sup> Robotic Grasping of Manipulation Competition - Picking in Clutter** May. 2024  
IEEE, IEEE RAS [RGMC 2024 🌐]
- 🏆 **KSME Student Creative Design Competition** Oct. 2021  
The Korean Society of Mechanical Engineers (KSME) [YouTube 🌐]
- 🏆 **Creative and Intelligent Robot Contest** Sep. 2021  
Daejeon Metropolitan City, Chungnam National University [YouTube 🌐]

## EXPERIENCE

- **CarbonSix** Oct. 2024 - Nov. 2024, Jun. 2026 - Jul. 2026  
Researcher Gangnam-daero, Seocho-gu, Seoul, Republic of Korea
  - Designed direct teaching device with gravity compensation function, and camera calibration device.
- **AIDIN ROBOTICS - Cobot Solution Team** May 2022 - Present  
Researcher Anyang-si, Gyeonggi-do, Republic of Korea
  - Designed grippers for shelf-picking solutions, contributing to efficient logistics automation.
- **Korea Institute of Industrial Technology (KITECH)** Oct 2021 - Dec 2021  
Research Student Ansan-si, Gyeonggi-do, Republic of Korea
  - Designed and developed control systems for a 2-DoF manipulator, enhancing its performance and accuracy.
- **Wall Climbing Car (WCC) [Undergraduate Project]** Dec 2020 - Oct 2021  
Team Leader Ansan-si, Gyeonggi-do, Republic of Korea
  - Designed a wall-climbing car's propeller frame and control system.

## PERSONAL PROJECTS

- **Learning from Human Corrections | Residual Learning** 2026  
[🌐] [🔗]
  - Trained a residual policy from human corrections to refine a frozen diffusion-based base policy for contact-rich robotic box insertion.
  - Evaluated at the Original position and an unseen backward target shift of 3 cm; the base policy used absolute-action demonstrations collected only at the Original position.

## • Scooping Grasping with Robotic Hand | Wrist F/T sensing, Compliance control

May 2026 - Present



- **Objective:** Enabling a rigid-fingertip robotic hand to scoop objects off the floor, a motion in which environmental contact would otherwise close a kinematic chain and damage the finger mechanism.
- Wrench measured in wrist is fed into a Cartesian impedance controller so that the manipulator responds compliantly to both wrist- and fingertip-level contacts, achieving safe interaction against external constraints without sacrificing the grasping force and positional accuracy.

## • Impedance Control with Doosan M0609

Jan. 2026 - Present



- **Objective:** Overcoming the limitations of position control in teleoperated manipulation, where contact-rich tasks and constrained environments cannot be handled reliably.
- Implemented a task-space impedance controller running on a 1 kHz real-time torque-control loop over the Doosan real-time API.
- Developed a momentum-based disturbance observer (De Luca formulation) for external-contact estimation, together with gravity, tool, and friction compensation.
- Demonstrated contact maintenance under continuous force and impact, and pivoting tasks in constrained environments, where position control fails.

## PROJECTS

### • Development of a K-Logistics Humanoid Robot Integrated with a High-Sensitivity Robotic Hand Based on a Multimodal AI Foundation Model

Sep 2025 - Present

Participating organizations: AIDIN ROBOTICS, SKKU, KETI, CJ Logistics

Republic of Korea

- **Research Objective:** Automation of contact-rich manipulation tasks using a humanoid robot equipped with a high-sensitivity robotic hand.
- Development of a teleoperation system for constructing multimodal datasets incorporating force information
- Design of force-control strategies and learning-based policies capable of handling contact-rich interactions

### • Development of Smart Vision System and All-in-One Universal Gripper for Multi-Variou Random Piece Picking

May 2022 - 2024

Participating organizations: SKKU, AIDIN ROBOTICS, KITECH, CJ Logistics

Republic of Korea

- **Research Objective:** Developing robotic picking system including a gripper and vision system for piece picking in a logistics environment.
- Designing an integrated gripper capable of various grasping strategy for a shelf environment among logistics environments
- Researching grasping strategy that can pick objects while avoiding external constraints in a shelf environment

## GRANT

- BK21 Research Encouragement Scholarship 
- Teaching Assistantship Scholarship
- Brain Hanyang, Academic Excellence Scholarship

Fall. 2024, and Spring. 2026

Spring. 2022, Fall. 2022

Spring. 2018

## REVIEW SERVICE

- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- Scientific Reports

## ADVISING EXPERIENCE

### • Research Mentoring

Proposed experimental methodology for a journal publication

- Proposed the experimental design for the T-MECH paper: *Design and Control of 2-DOF Differential Joint Driven by Soft Pneumatic Actuator for Versatile Soft Robotic Manipulator (recently accepted)*..

### • Undergraduate Student Mentoring

Mentored undergraduate students, leading to awards in university-level and national competitions

- Guided undergraduate students in robotics research and project development, resulting in awards at both university-level and national competitions.
- Mentees: Taehwan Yoon, Jeonghwan Park, Sihun Ryu, Hyunjun Jo, Kihyeok Kwon, Seongmin Kim, Seongmin Ryu.